Generative AI End-Term Project

Dual Text RAG and Multimodal RAG

STAI345 – Generative AI | Vijaybhoomi University

Research Incubation Handbook as the knowledge source

1. Problem Statement

The objective was to build and evaluate a local Retrieval-Augmented Generation (RAG) system using the Vijaybhoomi University Research Incubation Handbook. Two lightweight language models were compared in the same RAG pipeline. The project was then extended from text-only RAG to Multimodal RAG so that the system could use both handbook text and page images.

2. Approach

The project was completed in two phases:

1. Phase 1 – Build one common text RAG pipeline and run it with SmolLM2-135M and Llama 3.2-1B.
2. Evaluate both models using the same 20 questions, retrieval settings and evaluation code.
3. Analyze retrieval quality, answer quality, grounding, relevance and latency.
4. Phase 2 – Add a lightweight vision-language model (SmolVLM-256M) so handbook page images can also be used.
5. Expose the complete system through a Streamlit application.

3. Models and Main Components

Component	Used in the project
Small text LLM	SmolLM2-135M
Larger text LLM	Llama 3.2-1B Instruct, Q4_K_M GGUF
Multimodal model	SmolVLM-256M-Instruct, Q8_0 GGUF
Embeddings	Sentence-transformer based embeddings
Vector database	FAISS
Backend	Python modules for document processing, retrieval, generation and evaluation
Multimodal runtime	llama.cpp CLI
User interface	Streamlit

Both text models used the same handbook, embeddings, vector store, retrieval method and questions. This kept the model comparison controlled.

4. Directory Structure

```
generative_ai_endterm_rag/
├── app/
│   └── streamlit_app.py
├── src/
│   ├── config.py
│   ├── document.py
│   ├── embeddings.py
│   ├── vector_store.py
│   ├── llm.py
│   ├── rag.py
│   ├── evaluation.py
│   └── multimodal.py
├── data/
```

```
│   ├── university_handbook.pdf
│   └── evaluation_questions.json
├── notebooks/
│   ├── 01_document_processing.ipynb
│   ├── 02_dual_rag.ipynb
│   ├── 03_evaluation.ipynb
│   └── 04_multimodal_rag.ipynb
├── diagrams/
├── results/
├── requirements.txt
└── README.md
```

5. Phase 1 – Dual Text RAG

5.1 Document Processing

The Research Incubation Handbook PDF was loaded and converted into searchable text. The document was split into retrievable units, embedded, and stored in the vector database. For a user question, the retriever searches the vector store and returns the top relevant pages.

5.2 Retrieval and Generation

The same retrieval result was passed to each language model. The model was instructed to answer using the retrieved handbook context. This separates retrieval from generation and makes it possible to identify whether an error comes from retrieval or from the language model.

5.3 Evaluation Method

Twenty handbook-based questions were used for both models. Each model was evaluated independently using the same questions and the same retrieval configuration.

- Context Precision – measures how much of the retrieved context is relevant.
- Context Recall – measures whether the required source information was retrieved.
- MRR (Mean Reciprocal Rank) – measures how early the relevant page appears.
- Answer Similarity – compares the generated answer with the reference answer.
- Keyword Coverage – checks how many expected answer keywords were covered.
- Answer Correctness Proxy – lightweight overlap-based proxy for correctness.
- Groundedness Proxy – lightweight overlap-based proxy for support from retrieved text.
- Answer Relevancy Proxy – lightweight measure of answer relevance to the question.
- Latency – time taken to generate an answer.

The correctness, groundedness and relevancy measures are explicitly lightweight proxy metrics, not full human evaluation or a full LLM-judge/RAGAS faithfulness evaluation.

5.4 Phase 1 Results

Metric	SmolLM2-135M	Llama 3.2-1B Q4_K_M
Questions	20	20
Context Precision	0.2666	0.2666
Context Recall	0.8000	0.8000
MRR	0.6333	0.6333
Answer Similarity	0.2549	0.5021
Keyword Coverage	0.5483	0.7324
Answer Correctness Proxy	0.4016	0.6173
Groundedness Proxy	0.7551	0.8145
Answer Relevancy Proxy	0.8717	0.8324
Average Latency	24.479 s	29.969 s

5.5 Analysis of Phase 1

- Retrieval performance was identical for both models: Context Precision = 0.2666, Context Recall = 0.8000 and MRR = 0.6333.
- Llama 3.2-1B produced higher answer similarity, keyword coverage, correctness proxy and groundedness proxy.
- SmolLM2-135M was faster on average: 24.479 seconds compared with 29.969 seconds for Llama.
- The results show that the retrieval stage was not the main difference between the two models because both received the same retrieval results.
- Some failures came from noisy page-level retrieval. For example, cover/contact pages could be retrieved together with the relevant page.
- The smaller model also showed more repetition and incomplete answers on several questions, while the larger model generally produced more complete answers.

5.6 Model Comparison for This Experiment

For this specific 20-question experiment, Llama 3.2-1B gave stronger answer-quality results, while SmolLM2-135M had lower average latency. Therefore, the measured trade-off was: Llama 3.2-1B for better answer quality, and SmolLM2-135M for faster lightweight inference.

6. Phase 2 – Multimodal RAG

The text-only pipeline was extended so that a retrieved handbook page could also be rendered as an image and passed to a lightweight vision-language model. SmolVLM-256M-Instruct was used for this stage.

6. Retrieve relevant handbook pages using the existing text retriever.
7. Render the selected PDF page as a PNG image.
8. Send the retrieved text and the page image to SmolVLM.
9. Ask the model to answer using both the textual context and visual page content.
10. Record page retrieval, keyword coverage and latency.

6.1 Multimodal Evaluation

ID	Question	Expected Page	Retrieved Pages	Keyword Coverage	Expected Page Retrieved	Latency
MM01	What information is shown on page 3 of the Research Incubation Handbook?	3	1, 15	0.00	False	27.206 s
MM02	What grades and grade points are specified for the Research Incubation course?	11	6, 11	0.70	True	19.108 s

MM01 did not retrieve the expected page 3 and therefore had 0% keyword coverage. MM02 retrieved the expected page 11 and achieved 70% keyword coverage. The observed results show that multimodal generation cannot compensate for poor retrieval.

6.2 Multimodal Limitation Observed

The multimodal model sometimes produced incorrect or incomplete interpretations when the correct page was not retrieved. This confirms that a multimodal RAG system still depends strongly on retrieval quality. The current Phase

2 evaluation is a small demonstration set of two questions, so it should be treated as an initial evaluation rather than a statistically strong comparison.

7. Streamlit Application

The project was packaged into a simple Streamlit interface with three main sections:

- RAG Chat – select SmolLM2-135M or Llama 3.2-1B, enter a question, view the answer, retrieved pages and latency.
- Evaluation – view the saved evaluation results and compare the two text models.
- Multimodal RAG – submit a question and use the retrieved handbook page image together with text.

8. Failure Analysis and Root Causes

Observed Issue	Likely Root Cause	Effect
Irrelevant pages retrieved	Page-level retrieval and similar document text	Lower context precision
Relevant page sometimes missing	Embedding/retrieval mismatch	Lower recall and wrong downstream answer
Small model repetition	Limited model capacity and context handling	Lower answer similarity/correctness
Wrong multimodal answer	Expected page not retrieved or visual content not interpreted correctly	Low keyword coverage
Higher Llama latency	Larger model and CPU inference	Slower response time

9. Conclusion

The project successfully implemented a reproducible local dual-model RAG pipeline and extended it to a lightweight Multimodal RAG pipeline. On the 20-question Phase 1 evaluation, retrieval metrics were the same for both models, while Llama 3.2-1B achieved stronger answer-quality metrics and SmolLM2-135M was faster. Phase 2 demonstrated how retrieved handbook pages can be combined with page images, while also showing that retrieval quality remains critical. The Streamlit application provides a single interface for demonstrating the complete system.

10. Reproducibility

- Use the supplied requirements.txt and setup instructions.
- Place the Research Incubation Handbook in data/university_handbook.pdf.
- Run document processing and build the vector store.
- Run the two RAG models using the same evaluation questions.
- Run the evaluation script to regenerate CSV and JSON results.
- Run the multimodal pipeline after rendering the required handbook pages.
- Start Streamlit to demonstrate the complete system.

11. Project Output Files

- smollm_evaluation.json / .csv – 20-question SmolLM2 evaluation.
- llama_evaluation.json / .csv – 20-question Llama evaluation.
- smollm_summary.json and llama_summary.json – aggregate Phase 1 metrics.
- multimodal_evaluation.json – Phase 2 evaluation results.
- Architecture diagrams, notebooks, source code and Streamlit application.